

Table 3-135 lists the results of attempted access for each mode.

Table 3-135 Results of access to the Secure User and Non-secure Access Validation Control Register

Secure Privileged		Non-secure Privileged	User
Read	Write		
Data	Data	Undefined exception	Undefined exception

To access the Secure User and Non-secure Access Validation Control Register read or write CP15 with:

- Opcode_1 set to 0
- CRn set to c15
- CRm set to c9
- Opcode_2 set to 0.

For example:

MRC p15, 0, <Rd>, c15, c9, 0 ; Read Secure User and Non-secure Access Validation Control Register
MCR p15, 0, <Rd>, c15, c9, 0 ; Write Secure User and Non-secure Access Validation Control Register

3.2.51 c15, Performance Monitor Control Register

The purpose of the Performance Monitor Control Register is to control the operation of:

- the Cycle Counter Register
- the Count Register 0
- the Count Register 1.

Table 3-136 on page 3-134 lists the purpose of the individual bits in the register.

The Performance Monitor Control Register is:

- in CP15 c15
- a 32-bit read/write register common to Secure and Non-secure worlds
- accessible in User and Privileged modes.

Figure 3-73 shows the bit arrangement for the Performance Monitor Control Register.

31	28	27		20	19		12	11	10	9	8	7	6	5	4	3	2	1	0
SBZ/UNP			EvtCount0			EvtCount1			X	C	C	C	S	E	E	E	D	C	P
									R	C	R	R	B	C	C	C			E
										1	0	0	Z	C	1	0			

Figure 3-73 Performance Monitor Control Register format

Table 3-136 lists how the bit values correspond with the Performance Monitor Control Register.

Table 3-136 Performance Monitor Control Register bit functions

Bits	Field name	Function
[31:28]	-	UNP/SBZ.
[27:20]	EvtCount0	Identifies the source of events for Count Register 0. Table 3-137 on page 3-135 lists the values, functions and EVNTBUS bit position for Count Register 0. The reset value is 0.
[19:12]	EvtCount1	Identifies the source of events for Count Register 1. Table 3-137 on page 3-135 lists the values and the bit functions for Count Register 1. The reset value is 0.
[11]	X	Enable Export of the events to the event bus to an external monitoring block, such as the ETM to trace events: 0 = Export disabled, EVNTBUS held at 0x0, reset value 1 = Export enabled, EVNTBUS driven by the events.
[10]	CCR	Cycle Counter Register overflow flag: 0 = For reads No overflow, reset value. For writes No effect. 1 = For reads, overflow occurred. For writes Clear this bit.
[9]	CR1	Count Register 1 overflow flag: 0 = For reads No overflow, reset value. For writes No effect. 1 = For reads, overflow occurred. For writes Clear this bit.
[8]	CR0	Count Register 0 overflow flag: 0 = For reads No overflow, reset value. For writes No effect. 1 = For reads overflow occurred. For writes Clear this bit.
[7]	-	UNP/SBZ.
[6]	ECC	Used to enable and disable Cycle Counter interrupt reporting: 0 = Disable interrupt, reset value 1 = Enable interrupt.
[5]	EC1	Used to enable and disable Count Register 1 interrupt reporting: 0 = Disable interrupt, reset value 1 = Enable interrupt.
[4]	EC0	Used to enable and disable Count Register 0 interrupt reporting: 0 = Disable interrupt, reset value 1 = Enable interrupt.
[3]	D	Cycle count divider: 0 = Counts every processor clock cycle, reset value 1 = Counts every 64th processor clock cycle.

Table 3-136 Performance Monitor Control Register bit functions (continued)

Bits	Field name	Function
[2]	C	Cycle Counter Register Reset. Reset on write, Unpredictable on read: 0 = No action, reset value 1 = Reset the Cycle Counter Register to 0x0.
[1]	P	Count Register 1 and Count Register 0 Reset. Reset on write, Unpredictable on read: 0 = No action, reset value 1 = Reset both Count Registers to 0x0.
[0]	E	Enable all counters: 0 = All counters disabled, reset value 1 = All counters enabled.

The Performance Monitor Control Register:

- controls the events that Count Register 0 and Count Register 1 count
- indicates the counter that overflowed
- enables and disables the report of interrupts
- extends Cycle Count Register counting by six more bits, cycles between counter rollover = 2^{38}
- resets all counters to zero
- enables the entire performance monitoring mechanism.

Table 3-137 lists the events that can be monitored using the Performance Monitor Control Register.

Table 3-137 Performance monitoring events

EVNTBUS bit position	Event number	Event definition
-	0xFF	An increment each cycle.
-	0x26	Procedure return instruction executed and return address predicted incorrectly. The procedure return address was restored to the return stack following the prediction being identified as incorrect.
-	0x25	Procedure return instruction executed and return address predicted. The procedure return address was popped off the return stack and the core branched to this address.
-	0x24	Procedure return instruction executed. The procedure return address was popped off the return stack.
-	0x23	Procedure call instruction executed. The procedure return address was pushed on to the return stack.
-	0x22	If both ETMEXTOUT[0] and ETMEXTOUT[1] signals are asserted then the count is incremented by two. If either signal is asserted then the count increments by one.
-	0x21	ETMEXTOUT[1] signal was asserted for a cycle.
-	0x20	ETMEXTOUT[0] signal was asserted for a cycle.
[19]	0x12	Write Buffer drained because of a Data Synchronization Barrier operation or Strongly Ordered operation.

Table 3-137 Performance monitoring events (continued)

EVENTBUS bit position	Event number	Event definition
[18]	0x11	Stall because of a full Load Store Unit request queue. This event takes place each clock cycle when the condition is met. A high incidence of this event indicates the LSU is often waiting for transactions to complete on the external bus.
[17]	0x10	Explicit external data accesses, Data Cache linefills, Noncacheable, write-through.
[16]	0xF	Main TLB miss.
[15:14]	0xD	Software changed the PC. This event occurs any time the PC is changed by software and there is not a mode change. For example, a MOV instruction with PC as the destination triggers this event. Executing an SVC from User mode does not trigger this event, because it incurs a mode change. If EVENTBUS bit [15] is HIGH, two software PC changes occurred in this clock cycle and the count increments by two.
[13]	0xC	Data cache write-back. This event occurs once for each half line of four words that are written back from the cache.
[12]	0xB	Data cache miss. Does not include Cache Operations.
[11]	0xA	Data cache access. Does not include Cache Operations. This event occurs for each nonsequential access to a cache line, regardless of whether or not the location is cacheable.
[10]	0x9	Data cache access. Does not include Cache Operations. This event occurs for each nonsequential access to a cache line, for cacheable locations.
[9:8]	0x7	Instruction executed. If EVENTBUS bit [9] is HIGH, two instructions were executed in this clock cycle and the count is increments by two.
[7]	0x6	Branch mispredicted.
[6]	-	Reserved.
[5]	0x5	Branch instruction executed, branch might or might not have changed program flow.
[4]	0x4	Data MicroTLB miss.
[3]	0x3	Instruction MicroTLB miss.
[2]	0x2	Stall because of a data dependency. This event occurs every cycle when the condition is present.
[1]	0x1	Stall because instruction buffer cannot deliver an instruction. This can indicate an Instruction Cache miss or an Instruction MicroTLB miss. This event occurs every cycle when the condition is present.
[0]	0x0	Instruction cache miss.
Note This event counts all instruction cache misses, including any speculative access that would be a cache miss. If the instruction that caused a speculative access is not executed then there might not be a fetch from external memory. This can happen, for example, if the code branches round the instruction. This means that the value returned in this counter can be much larger than the number of external memory accesses caused by instruction cache misses.		
-	All other values	Reserved. Unpredictable behavior.

Access to the Performance Monitor Control Register in User mode depends on the V bit, see *c15, Secure User and Non-secure Access Validation Control Register* on page 3-132. The Performance Monitor Control Register is always accessible in Privileged modes. Table 3-138 lists the results of attempted access for each mode.

Table 3-138 Results of access to the Performance Monitor Control Register

V bit	Secure Privileged		Non-secure Privileged		User	
	Read	Write	Read	Write	Read	Write
0	Data	Data	Data	Data	Undefined exception	Undefined exception
1	Data	Data	Data	Data	Data	Data

To access the Performance Monitor Control Register read or write CP15 with:

- Opcode_1 set to 0
- CRn set to c15
- CRm set to c12
- Opcode_2 set to 0.

For example:

```
MRC p15, 0, <Rd>, c15, c12, 0    ; Read Performance Monitor Control Register
MCR p15, 0, <Rd>, c15, c12, 0    ; Write Performance Monitor Control Register
```

If this unit generates an interrupt, the processor asserts the pin **nPMUIRQ**. You can route this pin to an external interrupt controller for prioritization and masking. This is the only mechanism that signals this interrupt to the core. When asserted, this interrupt can only be cleared if bit 0 of the Performance Monitor Control Register is high.

There is a delay of three cycles between an enable of the counter and the start of the event counter. The information used to count events is taken from various pipeline stages. This means that the absolute counts recorded might vary because of pipeline effects. This has negligible effect except in cases where the counters are enabled for a very short time.

In addition to the two counters within the processor, most of the events that Table 3-137 on page 3-135 lists are available on an external bus, **EVNTBUS**. You can connect this bus to the ETM unit or other external trace hardware to enable the events to be monitored. If you do not want this functionality, set the X bit in the Performance Monitor Control Register to 0. In Debug state, the **EVNTBUS** is masked to zero.

3.2.52 c15, Cycle Counter Register

The purpose of the Cycle Counter Register is to count the core clock cycles.

The Cycle Counter Register:

- is in CP15 c15
- is a 32-bit read/write register common to Secure and Non-secure worlds
- counts up and can trigger an interrupt on overflow.

The Cycle Counter Register bits[31:0] contain the count value. The reset value is 0.

You can use this register in conjunction with the Performance Monitor Control Register and the two Counter Registers to provide a variety of useful metrics that enable you to optimize system performance.

Access to the Cycle Counter Register in User mode depends on the V bit, see *c15, Secure User and Non-secure Access Validation Control Register* on page 3-132. The Cycle Counter Register is always accessible in Privileged modes. Table 3-139 lists the results of attempted access for each mode.

Table 3-139 Results of access to the Cycle Counter Register

V bit	Secure Privileged		Non-secure Privileged		User	
	Read	Write	Read	Write	Read	Write
0	Data	Data	Data	Data	Undefined exception	Undefined exception
1	Data	Data	Data	Data	Data	Data

To access the Cycle Counter Register read or write CP15 with:

- Opcode_1 set to 0
- CRn set to c15
- CRm set to c12
- Opcode_2 set to 1.

For example:

```
MRC p15, 0, <Rd>, c15, c12, 1    ; Read Cycle Counter Register
MCR p15, 0, <Rd>, c15, c12, 1    ; Write Cycle Counter Register
```

The value in the Cycle Counter Register is zero at Reset.

You can use the Performance Monitor Control Register to set the Cycle Counter Register to zero.

You can use the Performance Monitor Control Register to configure the Cycle Counter Register to count every 64th clock cycle.

3.2.53 c15, Count Register 0

The purpose of the Count Register 0 is to count instances of an event that the Performance Monitor Control Register selects.

The Count Register 0:

- is in CP15 c15
- is a 32-bit read/write register common to Secure and Non-secure worlds
- counts up and can trigger an interrupt on overflow.

Count Register 0 bits [31:0] contain the count value. The reset value is 0.

You can use this register in conjunction with the Performance Monitor Control Register, the Cycle Count Register, and Count Register 1 to provide a variety of useful metrics that enable you to optimize system performance.

Note

- In Debug state the counter is disabled.
- When the core is in a mode where noninvasive debug is not permitted, set by **SPNIDEN** and the **SUNIDEN** bit, see *c1, Secure Debug Enable Register* on page 3-54, the processor does not count events.

Access to the Count Register 0 in User mode depends on the V bit, see *c15, Secure User and Non-secure Access Validation Control Register* on page 3-132. The Count Register 0 is always accessible in Privileged modes. Table 3-140 lists the results of attempted access for each mode.

Table 3-140 Results of access to the Count Register 0

V bit	Secure Privileged		Non-secure Privileged		User	
	Read	Write	Read	Write	Read	Write
0	Data	Data	Data	Data	Undefined exception	Undefined exception
1	Data	Data	Data	Data	Data	Data

To access Count Register 1 read or write CP15 with:

- Opcode_1 set to 0
- CRn set to c15
- CRm set to c12
- Opcode_2 set to 2.

For Example:

```
MRC p15, 0, <Rd>, c15, c12, 2    ; Read Count Register 0
MCR p15, 0, <Rd>, c15, c12, 2    ; Write Count Register 0
```

The value in Count Register 0 is 0 at Reset.

You can use the Performance Monitor Control Register to set Count Register 0 to zero.

3.2.54 c15, Count Register 1

The purpose of the Count Register 1 is to count instances of an event that the Performance Monitor Control Register selects.

The Count Register 1:

- is in CP15 c15
- is a 32-bit read/write register common to Secure and Non-secure worlds
- counts up and can trigger an interrupt on overflow.

Count Register 1 bits [31:0] contain the count value. The reset value is 0.

You can use this register in conjunction with the Performance Monitor Control Register, the Cycle Count Register, and Count Register 0 to provide a variety of useful metrics that enable you to optimize system performance.

———— Note ————

- In Debug state the counter is disabled.
- When the core is in a mode where non-invasive debug is not permitted, set by **SPNIDEN** and the SUNIDEN bit, see *c1, Secure Debug Enable Register* on page 3-54, the processor does not count events.

Access to the Count Register 1 in User mode depends on the V bit, see *c15, Secure User and Non-secure Access Validation Control Register* on page 3-132. The Count Register 1 is always accessible in Privileged modes. Table 3-141 lists the results of attempted access for each mode.

Table 3-141 Results of access to the Count Register 1

V bit	Secure Privileged		Non-secure Privileged		User	
	Read	Write	Read	Write	Read	Write
0	Data	Data	Data	Data	Undefined exception	Undefined exception
1	Data	Data	Data	Data	Data	Data

To access Count Register 1 read or write CP15 with:

- Opcode_1 set to 0
- CRn set to c15
- CRm set to c12
- Opcode_2 set to 3.

For example:

```
MRC p15, 0, <Rd>, c15, c12, 3    ; Read Count Register 1
MCR p15, 0, <Rd>, c15, c12, 3    ; Write Count Register 1
```

The value in Count Register 1 is 0 at Reset.

You can use the Performance Monitor Control Register to set Count Register 1 to zero.

3.2.55 c15, System Validation Counter Register

The purpose of the System Validation Counter Register is to count core clock cycles to trigger a system validation event.

The System Validation Counter Register is:

- in CP15 c15
- a 32 bit read/write register common to the Secure and Non-secure worlds
- accessible in User and Privileged modes.

The System Validation Counter Register consists of one 32-bit register that performs four functions. Table 3-142 lists the arrangement of the functions in this group. The reset value is 0.

Table 3-142 System validation counter register operations

CRn	Opcode_1	CRm	Opcode_2	R/W	Operation
c15	0	c12	1	R/W	Reset counter
			2	R/W	Interrupt counter
			3	R/W	Fast interrupt counter
			7	W	External debug request counter

The reset, interrupt, and fast interrupt counters are 32-bits wide. The external debug request counter is 6 bits wide. Figure 3-74 on page 3-141 shows the arrangement of bits for the external debug request counter.